# PhysiQuest VR


Project Team

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# Chapter 1

# Introduction

Physics, while foundational to understanding the world around us, often presents a significant learning challenge due to the abstract nature of its concepts. Students frequently struggle to grasp the real-world applications of theories learned in the classroom. This gap between theoretical knowledge and practical understanding can hinder their ability to fully appreciate the relevance of physics in everyday life. The purpose of this project is to address this disconnect by creating a gamified Virtual Re- ality (VR) educational system that immerses students in interactive environments where they can experience and apply physics concepts firsthand. By transforming abstract the- ories into tangible, real-world scenarios, the system aims to enhance comprehension and engagement, making learning more intuitive and enjoyable. Through the use of VR, stu- dents can explore and experiment with various physics phenomena in a controlled, simu- lated environment, bridging the gap between textbook learning and real-world application. This approach not only deepens students' understanding of physics but also sparks cu- riosity and encourages hands-on learning, all while providing an innovative solution to modern educational challenges.[**?** ].

Figure 1.1: PhysiQuest VR Logo

## 1.1 Problem Statement

Physics is often perceived as an abstract and challenging subject by students, where traditional teaching methods may not effectively convey complex concepts. Students struggle to connect theoretical knowledge with real-world applications, leading to disengagement and difficulty in understanding essential principles. This often results in gaps in comprehension and a lack of enthusiasm for the subject.

## 1.2 Scope

PhysiQuest VR will provides an immersive learning experience in physics for students aged 12-16, specifically those in Grade 6,7, and 8, as well as secondary education levels like Matric and O levels. In this project, we will cover fundamental topics in Mechanics, such as Newton's Laws, Kinematics, and Dynamics, focusing mainly on motion, forces and their relationship between mass, and acceleration. In addition to Newton's Laws, we will cover concepts from sound, like the Doppler Effect, as well as topics in Waves and Optics, such as total internal reflection and refraction. The project will comprise of multiple mini-games, each tackling a different physics con- cept. Players will be able to choose a mini-game from the main menu, depending on the concept they want to learn. There will be interactive simulations and challenges, each with their own unique storyline, designed to teach the associated theories in a practical manner.

## 1.3 Modules

In the development of PhysiQuest, various functional modules are designed to ensure an immersive and interactive learning experience. They are divided into two sub-modules .i.e the core modules and the mini-game modules. The core modules encompass the essential aspects like the mechanics, accuracy of the physical simulations, and user-friendliness of the system so that the players can interact with real-world physics concepts in a dynamic environment. The specialized mini-game modules focus on the specific physics concepts relative to each mini-game. Each module in this section, covers the physics topics that the mini-game will teach, and how they will be depicted. These functional requirements collectively support the system's educational objectives by combining interactive VR elements with physics learning, promoting deeper comprehension through hands-on simulations.

## 1.4 User Classes and Characteristics

The table outlines the various classes and that will interact with the VR Game and describes their roles. Each user has specific interactions tailored to their roles within the system.

Table 1.1: User Classes and Characteristics

| User class | Description |
|---|---|
| Students | As the primary users of the VR Game, the students, aged 12-16, will use the system to interact with mini-games that teach tough physics concepts like Newton's Laws, Doppler Effect, and refraction. The students will varying levels of prior knowledge of physics. The system should cater to different learning speeds, comprehensions, and abilities. |
| Teachers | The teachers will monitor student progress. They might review results from the mini-games, track student performance and select specific mini games based on the topics being taught in class. |
| Game Developers | Game Developers will build, maintain, and expand the VR educational system. They will work on the physics simulations, graphics, and game mechanics within each mini-game. They may interact with the system at infrastructure level, or to view, alter and debug the code, etc. |

# Chapter 2

# Project Requirements

The combination of the Use-Case diagram and detailed Use-Cases capture the key interactions in PhysiQuest VR. The detailed Use-Cases, functional and non-functional requirements illustrate the specific interaction patterns in all the mini-games, making sure that the player's actions are directly tied to the educational goals related to the physics concepts integrated in each game.

## 2.1 Use-cases Modelling

PhysiQuest VR involves interactive end-user applications where players (students) engage directly with the system to learn physics concepts through VR.

### 2.1.1 Use-Case Diagram

Below is the Use-Case Diagram illustrating how specific interactions will take place in PhysiQuest VR.

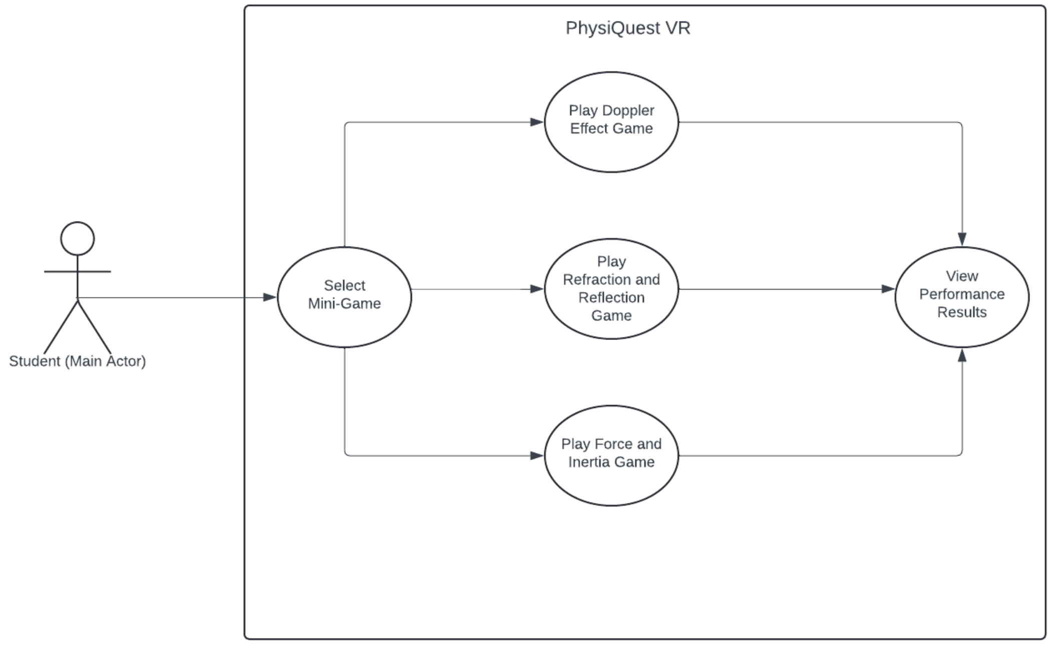


Figure 2.1: Use Case Diagram

### 2.1.2 Detailed Use-Cases

The detailed Use-Cases describe each Use-Case's detailed functionalities.

#### 2.1.2.1 Select Mini-Game

The following is the detailed use case for the Select Mini-Game use case.

Table 2.1: USC1: Select Mini-Game

| | |
|---|---|
| Use-Case Name | USC1: Select Mini-Game |
| Scope | PhysiQuest VR |
| Primary Actor | Student |
| Stakeholders and Interests | Student: Wants to play the mini-game that covers the physics topic they want to learn. |
| Precondition | The student is logged into the VR system. |
| Main Success Scenario | 1. The system presents a menu of available mini-games.<br>2. The student selects a mini-game to play.<br>3. The system loads the selected mini-game. |

#### 2.1.2.2 Play Doppler Effect Game

The following is the detailed use case for the Play Doppler Effect Game use case.

Table 2.2: USC2: Play Doppler Effect Game

| | |
|---|---|
| Use-Case Name | USC2: Play Doppler Effect Game |
| Scope | PhysiQuest VR |
| Primary Actor | Student |
| Stakeholders and Interests | Student: Wants to gain a deeper understanding of the Doppler Effect. |
| Precondition | The student has selected the Doppler Effect game from the main menu. |
| Main Success Scenario | 1. The student adjusts the speed of the spaceship based on changes in sound pitch.<br>2. The system provides real-time feedback on how changes in speed affect sound frequency.<br>3. The student successfully locates the stranded aliens based on pitch.<br>4. The system records the student's success and provides feedback. |

#### 2.1.2.3 Play Refraction and Reflection Game

The following is the detailed use case for the Play Refraction and Reflection Game use case.

Table 2.3: USC3: Play Refraction and Reflection Game

| | |
|---|---|
| Use-Case Name | USC3: Play Refraction and Reflection Game |
| Scope | PhysiQuest VR |
| Primary Actor | Student |
| Stakeholders and Interests | Student: Wants to learn more about the laws of refraction and reflection |
| Precondition | The student has selected the Refraction and Reflection game from the main menu. |
| Main Success Scenario | 1. The system presents the player with the game environment where they control an archer fish aiming to shoot at insects above the water.<br>2. The student adjusts the angle of the fish's shot, taking into account the bending of light as it passes between water and air (refraction).<br>3. The system provides visual feedback, showing how the light bends based on the angle chosen.<br>4. The student successfully adjusts the shot to hit the target, demonstrating an understanding of the refraction and reflection concepts.<br>5. The system evaluates the player's performance, including accuracy and the speed with which they adjusted the shot.<br>6. The system displays feedback on the player's understanding of light behavior and its real-world application. |

#### 2.1.2.4 Play Force and Inertia Game

The following is the detailed use case for the Play Force and Inertia Game use case.

Table 2.4: USC4: Play Force and Inertia Game

| | |
|---|---|
| Use-Case Name | USC4: Play Force and Inertia Game |
| Scope | PhysiQuest VR |
| Primary Actor | Student |
| Stakeholders and Interests | Student: Wants to grasp the concepts of Newton's Laws. |
| Precondition | The student has chosen the Force, Inertia, and Gravity game from the main menu. |
| Main Success Scenario | 1. The system places the player in a space-themed environment as an intergalactic delivery pilot.<br>2. The student navigates their spacecraft across different planetary terrains, experiencing changes in force and inertia based on the mass of their cargo and planetary gravity.<br>3. The system simulates real-time physics, including inertia, friction, and gravitational forces, affecting the movement of the spacecraft.<br>4. The student balances the speed, control, and load in such a way that the cargo get safely delivered across the terrains.<br>5. The system gives feedback helping the student manage forces and inertia, while notifying them of any damage to the cargo and the delivery time.<br>6. The system gives feedback to the user on the application of Newton's Laws of Motion to help them get a better understanding of the concept. |

#### 2.1.2.5 View Performance Results

The following is the detailed use case for the View Performance Results use case.

Table 2.5: USC5: View Performance Results

| | |
|---|---|
| Use-Case Name | USC5: View Performance Results |
| Scope | PhysiQuest VR |
| Primary Actor | Student |
| Stakeholders and Interests | Student: Wants to know how he/she performed in the game and how well they understood the concept. |
| Precondition | The student has completed a mini-game. |
| Main Success Scenario | 1. The system shows the student's performance results, including completion time, accuracy, and mastery of the physics concept.<br>2. The student reviews the performance result.<br>3. The system displays two options: Replay and Return to Main Menu.<br>4. The student selects from the two options. |

## 2.2 Functional Requirements

### 2.2.1 Core Modules

#### 2.2.1.1 Module 1: User Interaction

This module enables players to interact with the VR environment using VR controllers, making the experience more immersive. Players can perform various action within the VR space.

1. The system shall allow players to interact with the VR environment using VR controllers.
2. The system shall allow players to perform actions such as movement, object manipulation, and selection within the VR environment.

#### 2.2.1.2 Module 2: Game Mechanics

This module defines the objectives of each mini-game, ensuring that the missions and controls align with their respective physics concepts.

1. The system shall provide specific controls for each mini-game that align with the physics concepts being taught.

2. The system shall include clear objectives in each mini-game that support the educational goals.

#### 2.2.1.3 Module 3: Physics Simulations

This module focuses on the implementation of physics simulations realistically within the game environment. It ensures that the system is accurately imitating key physics concepts.

1. The system shall implement accurate simulations of physics concepts including forces, motion, sound waves, light behavior, and energy transformations

#### 2.2.1.4 Module 4: Dynamic Adjustments

This module ensures real-time adjustments to various physics parameters. The system can modify the scenario based on player interactions and inputs.

1. The system shall allow real-time adjustments to physics parameters (e.g., speed, force) based on player actions and game requirements.

#### 2.2.1.5 Module 5: Concept Integration

The module ensures that each game is built around a specific physics concept with clear demonstrations and applications within the environment.

1. The system shall ensure that each mini-game clearly integrates and demonstrates the targeted physics concept (e.g., Doppler Effect, refraction).

#### 2.2.1.6 Module 6: Narrative Integration

The module guarantees that the games have engaging story-lines, aligning the narrative with the educational goals to create an immersive learning experience.

1. The system shall ensure that the story-line and context of the mini-games are cohesive, engaging, and tied into the educational goals.

#### 2.2.1.7 Module 7: VR Interface

This module focuses on providing a user-friendly VR interface. It will allow players to browse through the menu and control the gameplay seamlessly.

1. The system shall provide an intuitive VR interface for navigating menus, accessing in-game information, and controlling game-play.

#### 2.2.1.8 Module 8: Visual and Audio Design

This module guarantees high-quality graphics and sound design, enhancing the edu- actional experience.

1. The system shall ensure high-quality graphics and sound design that enhance the immersive experience and support educational objectives.

### 2.2.2 Mini-Game Modules

#### 2.2.2.1 Module 1: Doppler Effect

This module focuses on teaching the Doppler Effect using a space-themed mini-game. Players must adjust the speed of their spaceship to locate stranded aliens based on changes in the pitch of their distress signals.

1. The system will demonstrate how changes in relative motion affect sound frequencies, highlighting the Doppler Effect.
2. The system shall allow players to adjust the speed of their spaceship to match the pitch of the signal, teaching them how motion affects sound perception.

#### 2.2.2.2 Module 2: Total Internal Reflection and Refraction

In this module, players control an archer fish hunting for insects, adjusting their shots based on how light bends when transitioning between water and air. The game introduces the concepts of refraction and total internal reflection.

1. The system will ensure that the game teaches players how light refracts when moving between two media and how critical angles can cause total internal reflection.
2. The system shall let players adjust the angle of their shot to account for the bending of light, simulating real-world optics.

#### 2.2.2.3 Module 3: Force, Inertia, and Gravity

This module explores Newton's Laws of Motion as players take on the role of intergalactic delivery pilots navigating different planetary terrains. The game covers the effects of force, gravity, and inertia on objects in motion.

1. The system will make sure that the game emphasizes the effects of gravity, friction, and inertia on motion, allowing players to experience real-world physics applications.

2. The system will allow players to balance speed, control, and load to successfully deliver cargo without losing control or damaging the goods.

## 2.3 Non-Functional Requirements

### 2.3.1 Usability Requirements

USE-1: The system shall provide and intuitive and user-friendly interface that allows players to scroll through the menus and control the game-play without requiring any prior experience.
USE-2: The system shall grant users access to help or tutorial sections to understand the game mechanics as well as the physics concepts included in the game.
USE-3: The system shall be designed to cater to students aged 12-16, ensuring that the complexity of the game matches the age group's abilities.

### 2.3.2 Compatability Requirements

COM-1: The system shall be compatible with Meta's Oculus Quest 2 and 3 devices, ensuring seamless integration with the VR platform.
COM-2: The system shall run on Unity with the Universal Render Pipeline (URP), supporting both Windows and macOS environments for development and testing.

### 2.3.3 Organizational Requirements

ORG-1: The system shall be developed using tools like Unity and Oculus SDK.
ORG-2: The system shall be coded in C-sharp.
ORG-3: The system shall use assets from the Unity Asset Store as well as assets created

by the developers of the PhysiQuest on 3D Modelling platforms like Blender.

## 2.4 Domain Model

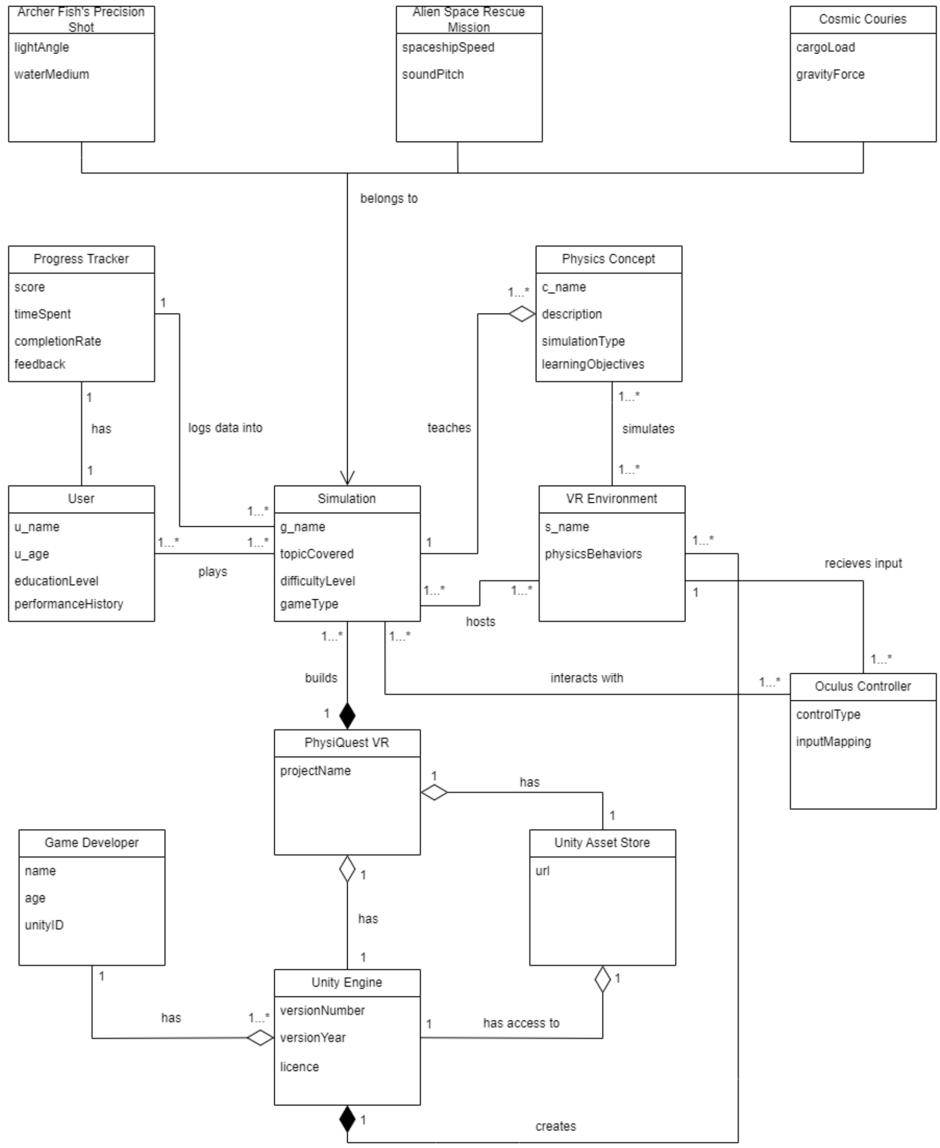


Figure 2.2: Domain Model

# Chapter 3

# System Overview

PhysiQuest VR is designed as a modular architecture, with core modules handling user interaction, game mechanics, and physics simulations. The core modules are tailored to each individual mini-game so that they align perfectly and accurately with the concepts the mini-games are focusing on. The architecture ensures that game mechanics are consistent, and that the core modules align with the learning concepts.

## 3.1 Architectural Design

The diagram below perfectly illustrates the relationship between the modules and how the sub-modules are connected with each other.

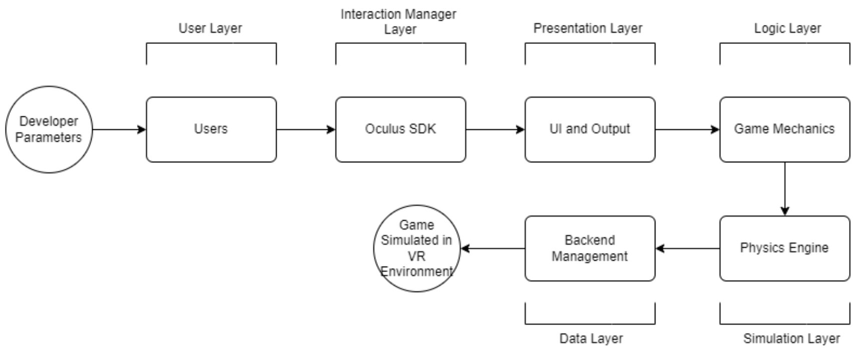


Figure 3.1: Architecture

## 3.2 Design Models

### 3.2.1 System Sequence Diagrams (SSD)

In order to model operations initiated between the user and the system, PhysiQuest VR, we have made use of system sequence diagrams.

#### 3.2.1.1 Main Game Flow

The SSD below illustrates the main process flow of the game, highlighting what the main interaction with the game will be like.

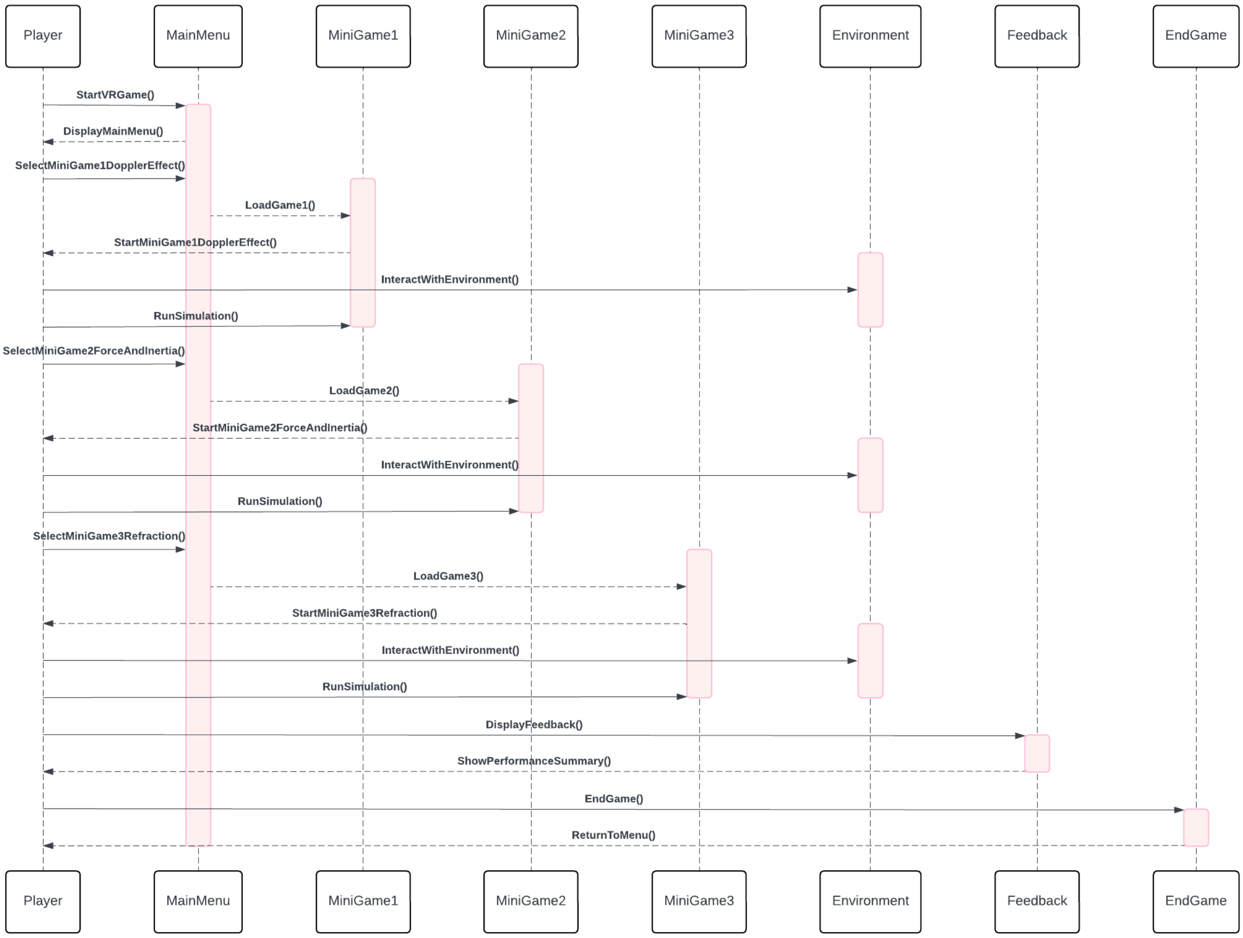


Figure 3.2: SSD Main Game Flow

#### 3.2.1.2 Doppler Effect Game

The following SSD illustrates the interaction between PhysiQuest VR and the Student while playing the mini game that covers the Doppler Effect concept.

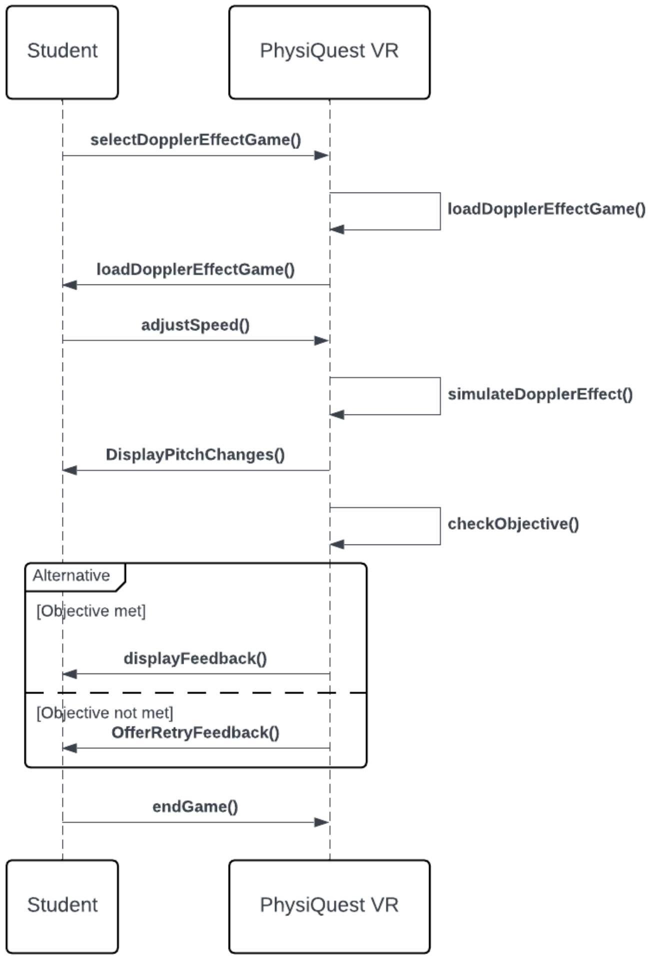


Figure 3.3: SSD Doppler Effect

#### 3.2.1.3 Refraction and Reflection Game

The following SSD illustrates the interaction between PhysiQuest VR and the Student while playing Refraction and Reflection Game.

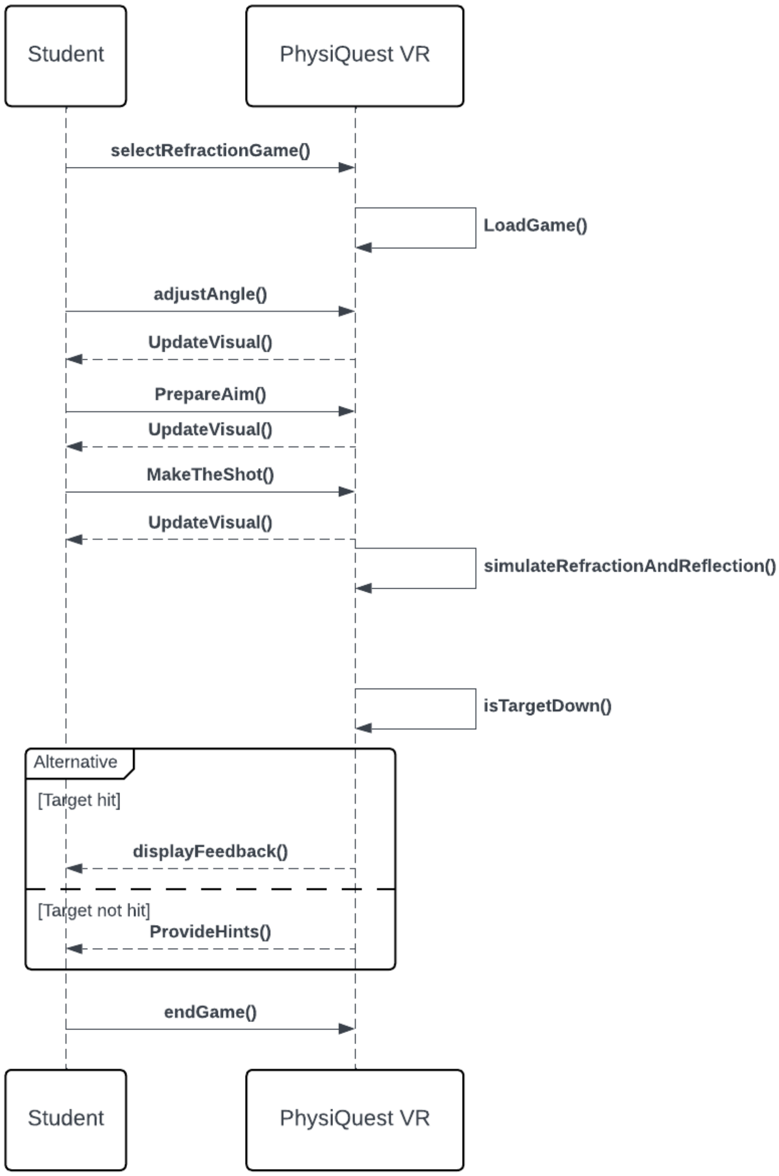


Figure 3.4: SSD Refraction and Reflection

#### 3.2.1.4 Force and Inertia Game

The following SSD illustrates the interaction that takes place when the Student is playing the Force and Inertia Game.

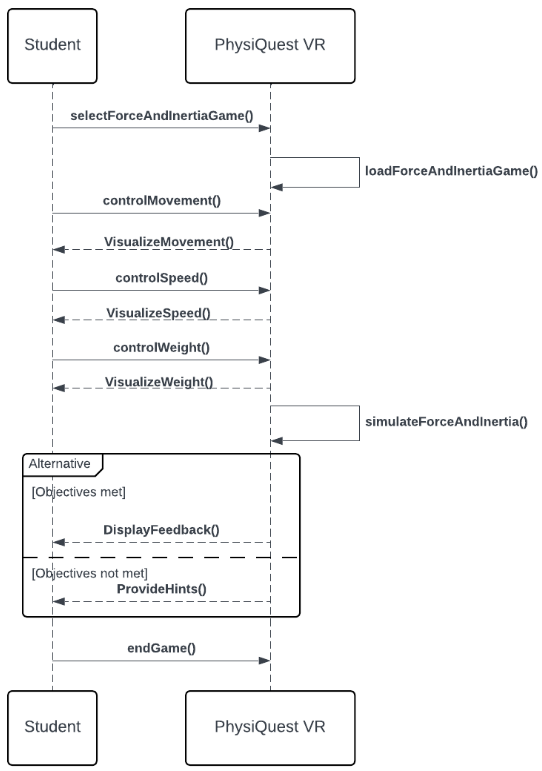


Figure 3.5: SSD Force and Inertia

**Design Models for Procedural Approach**

## 3.3 Activity Diagrams

### 3.3.1 Main Game Flow

The Activity diagram below showcases the main flow of the overall game.

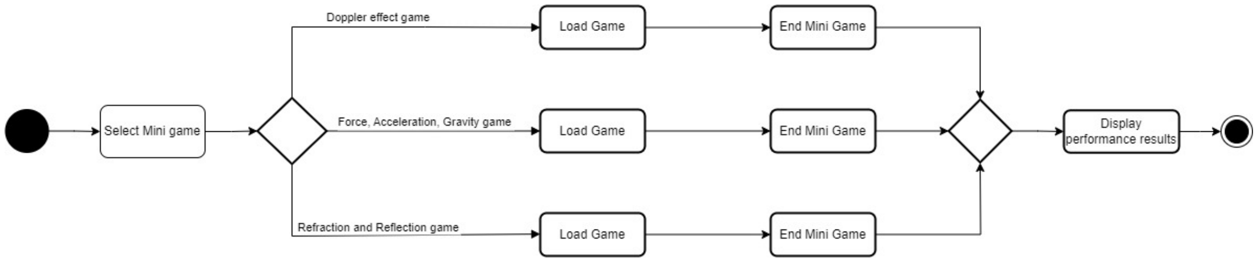


Figure 3.6: Main Game Flow Activity Diagram

### 3.3.2 Doppler Effect Game

The figure below shows the set of activities within the Doppler Effect mini game

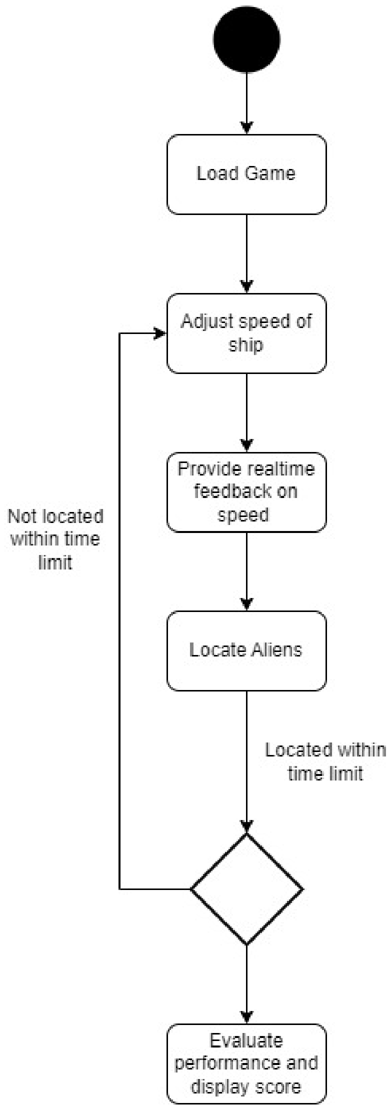


Figure 3.7: Doppler Effect Game Activity Diagram

### 3.3.3 Refraction and Reflection Game

The diagram below shows the activities that will be performed in the Reflection and Refraction mini game.

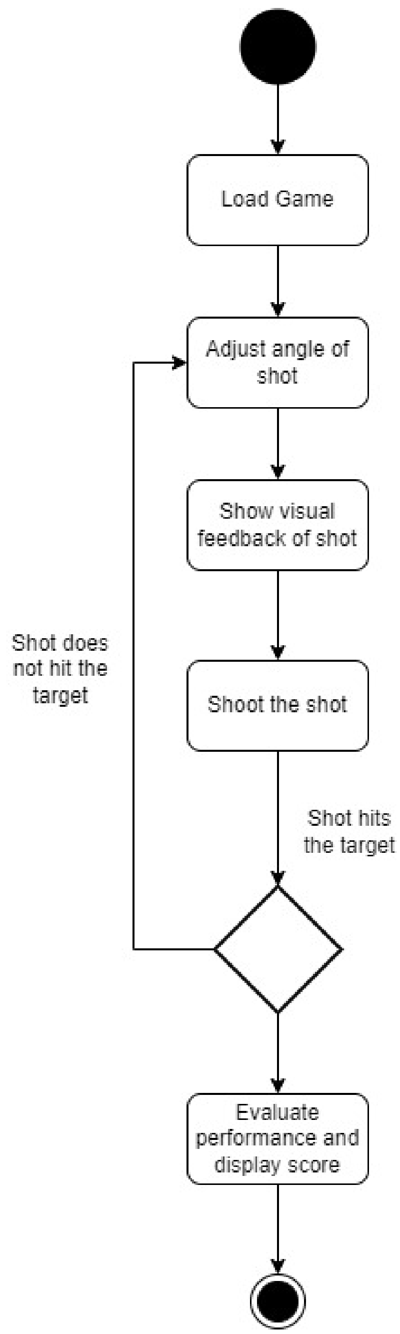


Figure 3.8: Reflection and Refraction Activity Diagram

### 3.3.4 Force and Inertia Game

The diagram below illustrates the main activities of the force, inertia, and gravity mini game.

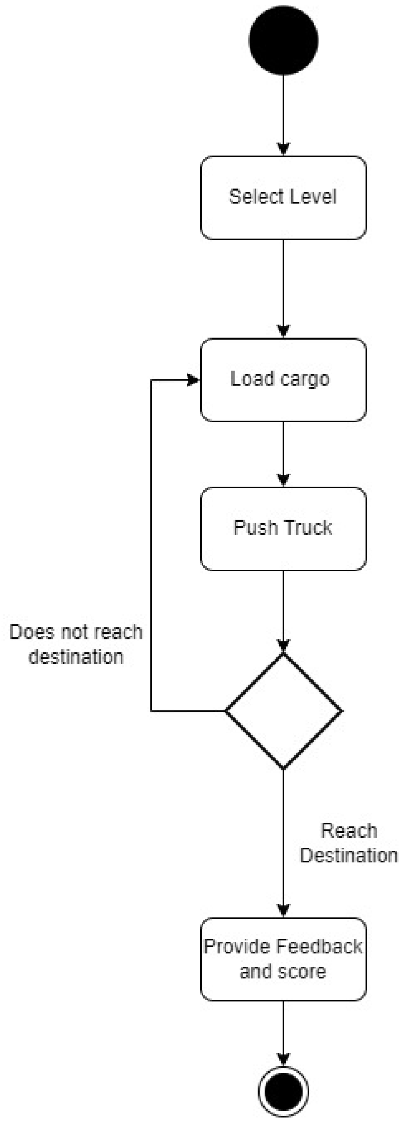


Figure 3.9: Force and Inertia Activity Diagram

## 3.4 Data Design

The data design for PhysiQuest VR is focused on structuring and organizing the necessary information for delivering an immersive and educational experience through physics-based mini-games. the various data entities that the system handles are user information, game progress, performance metrics like time taken in order to successfully complete a mini-game and accuracy when making decisions during the game, etc. These entities will need to be efficiently stored and managed.

### 3.4.1 Conversion to Data Structures

The main types of data include:

1. User Data: This includes the information about students, such as user IDs, progress, performance history, and achievements, which is stored persistently.
2. Game Data: This covers all the data specific to each mini-game such as object positions (spaceship in the Doppler Effect game or the archer fish in the Refraction game), game state (active or completed), and interaction variables (speed, angle, force).
3. Physics Simulation Data: This includes the data that gets temporarily stored during gameplay including speed, acceleration, refraction angles, etc, which will get processed dynamically.
4. Performance Data: This entity keeps track of how well the students perform in each game and is an essential part of the system as the data will be specific to each user and help them determine how well they've comprehended the concept being taught in the mini-game. The data includes metrics like: time taken to complete the mission in each game, accuracy, and game specific metrics (e.g., how well they adjusted forces or angles).

These information domains will be converted into data structures in the following way:

1. Arrays: Used to store multiple instances of game-related data (e.g., levels completed or user scores).
2. Classes and Objects: Each user, mini-game, or physics simulation is represented by an object containing attributes such as user ID, game status, and simulation parameters (force, inertia, speed, angle).

### 3.4.2 Databases and Data Storage

PhysiQuest VR will make use of databases for storage. An MSSQL database will be used to store user profiles, game progress, and performance metrics:

1. Users Table: Store User ID, Name, Age and Progress.

2. Game Sessions Table: Stores information like mini-game type, start time, end time, and completion status.

3. Performance Table: For tracking scores, completion times, and feedback for each game session.

The design ensures efficient storage and processing, allowing for real-time simulations and persistent progress tracking.

# Bibliography